SECTION PROFILES, CONCATENATION, AND SLAB HIERARCHIES FOR RANDOM TOEPLITZ AND HANKEL MOMENT POLYTOPES

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ABSTRACT. The moment polytopes of the limiting spectral laws γ_T and γ_H of random symmetric Toeplitz and Hankel matrices are studied through their sections by the initial coordinate. The resulting section profile is multiplicative under concatenation of double-occurrence words, invariant under ear insertion at every position, and flattens to a constant under a nesting pair. These three rules give an exact calculus: volumes of concatenations are computed by integrating products of profiles, insertions into a gap are computed against an explicit gap marginal, and the volume of a fully crossing r-pair word is $2/(r+1)$ in closed form. Consequences include supermultiplicativity of pairing volumes and of the primitive core masses, an exact cluster equation exhibiting the crossing correction to the Catalan equation, and closed-form extremal families: the reduced Toeplitz cores of at least five pairs attain $4/9$ exactly on one dihedral orbit for each size, and their parity-mixing counterparts of at least seven pairs attain $1/4$. All the slab constants appearing in this circle of problems – $2/3$, $1/2$, $2/5$, $1/3$ and the conjectural $2/7$ – are shown to be the single family $2/(j+2)$, with the exceptional system of the fully crossing word attaining each value. Finally, exact moment data through order twenty-two gives a four-moment certificate that γ_H is not unimodal, shows no obstruction to the unimodality of γ_T, and supports a conjectural description of γ_T as a scale mixture of semicircle laws.

1. INTRODUCTION

Let $(a_j)_{j\geq 0}$ be independent, identically distributed real random variables with mean zero and unit variance, and let T_n and H_n be the symmetric Toeplitz and Hankel matrices they generate. The empirical spectral distributions of $n^{-1/2}T_n$ and $n^{-1/2}H_n$ converge almost surely to deterministic symmetric laws γ_T and γ_H [1, 3], whose even moments are sums of volumes of pairing-indexed polytopes: for a pair partition π of $\{0,\dots,2k-1\}$ one forms

$$\mathcal{P}_T(\pi) = \big\{(x_0,t)\in\mathbb{R}^{k+1} : 0\leq x_0+S_\ell(t)\leq 1,\ 0\leq \ell<2k\big\},$$

with S_ℓ the signed partial sums of the increments attached to the pairs, and $m_{2k}(\gamma_T)=\sum_\pi p_T(\pi)$ with $p_T(\pi)=\operatorname{vol}\mathcal{P}_T(\pi)$; the Hankel law is the same sum restricted to parity-mixing pairings, with alternating forms. No closed-form identification of either law is known.

The companion analysis in [6] exposes a stable combinatorial skeleton behind these sums: repeated deletion of adjacent pairs ("ears") terminates in a canonical crossing core, volumes depend only on the core, and the resulting Catalan transform is exactly invertible. That analysis works with the volumes themselves. The present paper works one level lower, with the function

$$f_\pi(x) \;=\; \operatorname{vol}_k\big\{t:\ 0\leq x+S_\ell(t)\leq 1,\ 0\leq \ell<2k\big\},\qquad x\in[0,1],$$

the volume of the section of $\mathcal{P}_T(\pi)$ at initial coordinate x, so that $p_T(\pi)=\int_0^1 f_\pi$. Sections are finer than volumes: they are not invariant under rotation of the cyclic word, since rotating moves the base point. What they gain is a calculus. Section 2 proves that profiles multiply under concatenation, are unchanged by inserting an ear at any position, and collapse to a constant

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when a word is nested inside a single pair, and derives an exact formula for insertion into a gap. Section 3 computes the profile of the fully crossing word in closed form. These tools then do concrete work: supermultiplicativity of volumes and of the primitive core masses (Section 4), an exact cluster equation for the profile generating function (Section 5), extremal families realising 4/9 and 1/4 (Section 6), and the identification of every slab constant in the subject as one family (Section 7). Section 8 turns to the limiting laws themselves, where exact moments through order twenty-two give a short certificate of non-unimodality for γ_H and evidence for a semicircular-mixture description of γ_T.

Throughout, references to numbered results of the form "Theorem 4.1", "Lemma 8.1" and so on are to [6]. All computations described as exact were carried out in rational arithmetic; the verification scripts are available with the source package.

2. Concatenation and supermultiplicativity

For double-occurrence words w_1, w_2 on disjoint alphabets, write w_1w_2 for the concatenation, read as a cyclic word.

Lemma 2.1 (Profile factorization)**.** *For a pairing π let*

$$f_\pi(x) \;=\; \mathrm{vol}_k\{t:\ 0 \le x + S_\ell(t) \le 1,\ 0 \le \ell < 2k\}$$

be the x_0-section volume of $\mathcal{P}_T(\pi)$, so that $p_T(\pi) = \int_0^1 f_\pi$. Then

$$f_{w_1w_2}(x) \;=\; f_{w_1}(x)\, f_{w_2}(x), \qquad 0 \le x \le 1.$$

Moreover each f_π is symmetric about $x = \frac{1}{2}$ and log-concave.

Proof. Every pair contributes $+t_c$ and $-t_c$ to the walk, so after the last slot of w_1 the walk has returned to x_0, and the slot forms of the w_2-part of the concatenation involve only the w_2-variables. The constraint system therefore splits into the w_1-system and the w_2-system, coupled only through the common x_0; Fubini in the two independent blocks of t-variables gives the product formula.

Symmetry: the involution $(x_0, t) \mapsto (1 - x_0, -t)$ maps every slot form W_ℓ to $1 - W_\ell$ and hence preserves $\mathcal{P}_T(\pi)$; it sends the section at x to the section at $1 - x$.

Log-concavity: sections of a convex body by parallel hyperplanes have $\frac{1}{k}$-concave, in particular log-concave, volumes, by Brunn–Minkowski. □

Lemma 2.2 (Supermultiplicativity)**.** *For double-occurrence words on disjoint alphabets,*

$$p_T(w_1w_2) \;\ge\; p_T(w_1)\, p_T(w_2).$$

The same holds for p_H when w_1, w_2 and hence w_1w_2 are parity mixing.

Proof. By Lemma 2.1, $p_T(w_1w_2) = \int_0^1 f_{w_1} f_{w_2}$. Both factors are log-concave, hence unimodal, and symmetric about $\frac{1}{2}$; each is therefore a nondecreasing function of $-|x - \frac{1}{2}|$. Two nondecreasing functions of the same variable are nonnegatively correlated (Chebyshev's correlation inequality), so $\int_0^1 f_{w_1} f_{w_2} \ge \int_0^1 f_{w_1} \int_0^1 f_{w_2}$. The Hankel statement follows from Theorem 4.1, since concatenation of parity-mixing words along even cut points is parity mixing. □

Remark 2.3. Equality in Chebyshev's inequality forces one profile to be constant, and no nonempty word has constant profile (in every computed example $f_\pi(0) < f_\pi(\frac{1}{2})$; at $x_0 = 0$ the section satisfies the extra constraint $t_c \ge 0$ for the first opened pair), so the inequality is strict for nonempty words in all examples; we do not need the sharp equality classification here.

The lemma explains the shape of the extremal hierarchy: the unique four-pair Toeplitz maximizer 12123434 is the concatenation of two copies of the two-pair maximizer, with $p_T = \frac{9}{20} \ge (\frac{2}{3})^2 = \frac{4}{9}$, and the six-pair parity-mixing maximum $\frac{23}{90}$ is attained by the concatenation of two

three-pair cores of volume $\frac{1}{2}$, with $\frac{23}{90} \geq \frac{1}{4}$. Iterating the lemma on cube-cycle cores gives persistent-volume families with an exact lower bound $\prod_i 2/(r_i+1)$.

Lemma 2.4 (Cube-cycle profile)**.** *Let $w_r = 12\cdots r\,12\cdots r$ be the fully crossing r-pair word. Then its section profile is*

$$f_{w_r}(x) \;=\; \int_0^1 \min(x+y,\,2-x-y,\,1)^{\,r-1}\,dy \;=\; \frac{1-x^r}{r} + \frac{1-(1-x)^r}{r},$$

and consequently

$$p_T(w_r) \;=\; \int_0^1 f_{w_r} \;=\; \frac{2}{r+1}.$$

Proof. The slot forms of w_r are the prefix sums $x_0+t_1+\cdots+t_j$ $(1\leq j\leq r)$ and the suffix sums $x_0+t_{j+1}+\cdots+t_r$ $(1\leq j\leq r-1)$. Substitute $u_j = x+t_1+\cdots+t_j$; the map $t\mapsto u$ is unimodular, the prefix constraints become $u_j\in[0,1]$, and the suffix constraints become $u_r+x-u_j\in[0,1]$ for $j<r$. For fixed $u_r=y$, each of $u_1,\ldots,u_{r-1}$ ranges independently over $[0,1]\cap[x+y-1,\,x+y]$, an interval of length $\min(x+y,\,2-x-y,\,1)$. Integrating over $y\in[0,1]$ gives the first formula; splitting at $x+y=1$ gives the closed form. For the volume, with $s=x+y$ the sum of two independent uniforms,

$$p_T(w_r) = \int_0^2 \min(s,2-s,1)^{r-1}\,\big(1-|s-1|\big)\,ds = 2\int_0^1 s^{r-1}\cdot s\,ds = \frac{2}{r+1}. \qquad \square$$

Combining Lemmas 2.2 and 2.4: for any concatenation of cube cycles of sizes $r_1,\ldots,r_m$,

$$p_T \;\geq\; \prod_{i=1}^{m} \frac{2}{r_i+1},$$

an explicit persistent-volume family with, by Lemma 2.1, exactly computable volumes $\int_0^1 \prod_i f_{w_{r_i}}$.

Lemma 2.5 (Profile calculus)**.** *Let π be a pairing with section profile f_π as in Lemma 2.1.*

(i) *(Ear neutrality.) Inserting an ear into any gap of π leaves the profile unchanged; hence $f_\pi = f_{\mathrm{core}(\pi)}$ pointwise, not only after integration.*

(ii) *(Flattening.) If w is a complete word and a a new letter, then $f_{awa}(x) = p_T(w)$ for every $x\in[0,1]$: nesting under a single pair flattens the profile to the constant $p_T(w)$.*

(iii) *(Gap marginals.) For a word C and a gap g, let $\rho_{C,g}$ be the pushforward to $[0,1]$ of Lebesgue measure on $\mathcal{P}_T(C)$ under the slot form W_g. Then $\rho_{C,g}$ has total mass $p_T(C)$, is symmetric about $\frac{1}{2}$ and log-concave, and for every complete word w on a disjoint alphabet inserted contiguously at gap g,*

$$p_T\big(C[w@g]\big) \;=\; \int_0^1 f_w(u)\,\rho_{C,g}(du) \;\geq\; p_T(C)\,p_T(w).$$

Proof. (i) An ear inserted at a gap where the walk sits at W_g contributes one variable t constrained only by $W_g+t\in[0,1]$, which is an interval of length one for every admissible W_g; integrating it out gives the factor 1 at every fixed x_0. This is the argument of Theorem 3.4 read at fixed x_0.

(ii) Opening a moves the walk to $u=x+t_a$, which ranges over $[0,1]$ with unit Jacobian; the inner word runs from u and returns to u, and closing a returns to $x\in[0,1]$. Hence $f_{awa}(x)=\int_0^1 f_w(u)\,du = p_T(w)$.

(iii) Conditioning on the variables of C, the inserted word starts and ends at the position W_g and contributes $f_w(W_g)$; integrating over $\mathcal{P}_T(C)$ gives the formula. The involution $(x_0,t)\mapsto(1-x_0,-t)$ preserves $\mathcal{P}_T(C)$ and maps W_g to $1-W_g$, so $\rho_{C,g}$ is symmetric; it is log-concave as

the marginal of the indicator of a convex body (Prékopa). Both f_w and the density of $\rho_{C,g}$ are then nondecreasing functions of $-|u-\frac{1}{2}|$, and Chebyshev's correlation inequality on $[0,1]$ gives the bound. □

For cube cycles the gap marginals are not merely explicit; they are the profile itself.

Corollary 2.6 (Self-similarity of cycle gaps)**.** *For every gap g of the fully crossing word w_r,*

$$\rho_{w_r,g} \;=\; f_{w_r},$$

so for any complete word w on a disjoint alphabet,

$$p_T\big(w_r[w@g]\big) \;=\; \int_0^1 f_{w_r} f_w,$$

independently of which gap is used.

Proof. In the coordinates of Lemma 2.4 the slot form at an opener gap is u_j, which for fixed $u_r = y$ ranges over an interval of length $\min(x+y, 2-x-y, 1)$ together with the other $r-2$ free prefix coordinates. Writing $s = x+y$, whose density is $1-|s-1|$ on $[0,2]$, the marginal is $\rho(v) = \int_v^{v+1} \min(s, 2-s, 1)^{\,r-2}(1-|s-1|)\,ds$. On $[0,2]$ one has $1-|s-1| = \min(s, 2-s, 1)$, so the integrand is $\min(s, 2-s, 1)^{\,r-1}$ and $\rho = f_{w_r}$ by Lemma 2.4. The rest is Lemma 2.5(iii). □

At $r=2$ this reads $f_{abab}(v) = \frac{1}{2} + v(1-v) = \frac{3}{4} - (v-\frac{1}{2})^2$, so the four-pair extremal volume is an L^2 norm,

$$\frac{9}{20} \;=\; \int_0^1 f_{abab}(v)^2\,dv,$$

consistent with $f_{12123434} = f_{abab}^2$ and $p = \int f$. Likewise $\int_0^1 f_{w_3} f_{abab} = \frac{61}{180}$ and $\int_0^1 f_{w_4} f_{abab} = \frac{19}{70}$, matching the measured insertion values at every gap of *abcabc* and *abcdabcd* respectively, and $\int_0^1 f_{w_3}^2 = \frac{23}{90}$ is the six-pair parity-mixing maximum.

The affine equivalence of Theorem 4.1 fixes the initial coordinate x_0 and acts unimodularly on the fiber, so it identifies sections and not only volumes: for every parity-mixing pairing the Hankel and Toeplitz profiles are equal pointwise, $f_\pi^H(x) = f_\pi^T(x)$ for all $x \in [0,1]$. Every profile statement in this section therefore transfers verbatim to the Hankel ensemble on the parity-mixing class.

Part (iii) contains Lemma 2.2 (the seam gap) and shows that the paper's inflation structure extends from ears to arbitrary contiguous insertions, with an exact formula. For cycle cores the gap marginals coincide across gaps; for example every gap of *abab* receives *cdcd* with volume $\frac{9}{20}$ and every gap of *abcabc* receives *dede* with volume $\frac{61}{180}$, while the path-of-squares core *abacbdcd* has two gap classes, receiving $efef$ with volumes $\frac{1}{4}$ and $\frac{313}{1260}$. Since ears are profile-neutral, the Catalan inflation of Theorem 6.1 holds at the level of profiles: $\sum_\pi f_\pi(x) = \mathrm{Cat}_k + \sum_r \binom{2k}{k-r} K_r(x)$ with $K_r(x) = \sum_{\rho \in \mathcal{R}_{T,r}} f_\rho(x)$, and the generating function identities of Theorem 6.4 hold for each fixed x.

Lemma 2.7 (Supermultiplicativity of the primitive masses)**.** *For all $r, s \geq 2$,*

$$\kappa_{r+s}^T \;\geq\; \kappa_r^T\,\kappa_s^T, \qquad \kappa_{r+s}^H \;\geq\; \kappa_r^H\,\kappa_s^H.$$

In particular $\lim_r (\kappa_r^E)^{1/r}$ exists in $(0,\infty]$ for both ensembles.

Proof. Fix cyclic positions $1,\dots,2(r+s)$. For a reduced r-pair word w_1 on the first $2r$ positions and a reduced s-pair word w_2 on the rest, on disjoint alphabets, the concatenation $w_1 w_2$ is again reduced: linear adjacencies inside each block are a subset of that block's cyclic adjacencies, and the two seam adjacencies join letters of different alphabets. The map $(w_1, w_2) \mapsto w_1 w_2$ is

injective, since the concatenation determines its restriction to the first $2r$ positions. Therefore, using Lemma 2.2 termwise,

$$\kappa_{r+s} \;\geq\; \sum_{(w_1,w_2)} p(w_1 w_2) \;\geq\; \Big(\sum_{w_1} p(w_1)\Big)\Big(\sum_{w_2} p(w_2)\Big) = \kappa_r\,\kappa_s.$$

For the Hankel case both blocks have even length, so parity mixing is preserved and Theorem 4.1 applies termwise. Existence of the limit is Fekete's lemma applied to $\log \kappa_r$. □

Remark 2.8. The exact values are far above the bound ($\kappa_8 = \frac{687080}{9}$ against $\kappa_4^2 = \frac{31684}{225}$), reflecting that only cut-separated concatenations are counted. The data suggest the sharper asymptotic $\kappa_{r+1}/\big((r+1)\kappa_r\big) \to 2\rho^2$ with ρ the Sen–Virág constant: the exact ratios 1.4833, 1.4213, 1.4002, 1.3899, 1.3844 have geometrically shrinking decrements and extrapolate to 1.3734, within 10^{-3} of $2\rho^2 = 1.37414$. Non-separated interleavings satisfy no such inequality: 12341234 interleaves 1212 and 3434 and has $p_T = \frac{2}{5} < \frac{4}{9} = p_T(1212)\,p_T(3434)$ - simultaneously open pairs couple the corridor constraints destructively. So the binomial enhancement of the lemma is false, and the finer $r!\gamma^r$ scaling must come from the boundary-local structure of the Catalan transform rather than from correlation inequalities.

Proposition 2.9 (Cluster equation)**.** *Let $F(z;x) = \sum_k F_k(x) z^k$ with $F_k(x) = \sum_{\pi \in P_2(2k)} f_\pi(x)$ the profile sum over all pairings, read as linear words, and let $J(z;x)$ be the same sum restricted to crossing-indecomposable words (words that are neither a nontrivial concatenation of complete blocks nor of the nested form awa). Then, as formal power series,*

$$F(z;x) \;=\; \frac{1}{1 - z\,M(z) - J(z;x)}, \qquad M(z) \;=\; \int_0^1 F(z;x)\,dx,$$

where M is the even-moment generating function of γ_T. With $J \equiv 0$ this is the Catalan equation $C = 1/(1 - zC)$, so J is precisely the crossing correction to the semicircle law.

Proof. Every nonempty linear word factors uniquely as a concatenation of indecomposable complete words, and profiles multiply over concatenation (Lemma 2.1), so $F = 1/(1 - I)$ with I the indecomposable profile sum. An indecomposable word is either of the form awa with w complete - by the flattening rule its profile is the constant $p_T(w)$, and summing over w and the label of the enclosing pair gives the term $z\,M(z)$ - or crossing-indecomposable, giving J. Averaging the first identity over $x \in [0,1]$ gives the second. □

The identity was verified exactly through order z^4 at $x = \frac{1}{2}$: series inversion of the computed $F_k(\frac{1}{2}) = 1, 1, \frac{11}{4}, \frac{47}{4}, \frac{12823}{192}$ yields $J_k(\frac{1}{2}) = 0, \frac{3}{4}, \frac{55}{12}, \frac{6139}{192}$, matching the direct enumeration of crossing-indecomposable words. Iterating the same decomposition inside the gaps of crossing-indecomposable words expands J over gap-irreducible atoms with gaps refilled by F evaluated at the gap positions; the gap-marginal formula of Lemma 2.5(iii) is the first moment of this expansion. Note that section profiles are not invariant under rotation of the cyclic word - rotating moves the basepoint - so all sums here are over linear words; only the integrated volumes are rotation invariant.

The calculus closes on a class. Lemmas 2.1, 2.5(ii) and 2.4 together are an algorithm on the class of words generated from fully crossing cycles by concatenation and by nesting under a single pair:

$$W \;::=\; w_r \;\big|\; WW \;\big|\; aWa.$$

Proposition 2.10 (Closed evaluation)**.** *For every word in this class the section profile is a polynomial in x, computed from*

$$f_{w_r}(x) = \frac{1-x^r}{r} + \frac{1-(1-x)^r}{r}, \qquad f_{UV} = f_U f_V, \qquad f_{aUa} \equiv \int_0^1 f_U,$$

and p_T is the integral of that polynomial. Every such volume is therefore an explicit rational number, obtained without any polytope computation.

The class is larger than its grammar suggests. By Corollary 2.6 the gap marginal of a cycle is its own profile, so inserting a word w into any gap of w_r has volume $\int_0^1 f_{w_r} f_w$ – which is exactly the concatenation formula. Insertion into a cycle is therefore volume-equivalent to concatenation, and every word obtained by hanging cycles off the gaps of other cycles already has its volume realised inside the grammar. This is why the gap-independence observed for *abab*, *abcabc* and *abcdabcd* was not a coincidence.

Reducedness needs one remark. Read cyclically, aWa places the two endpoints of a at positions 0 and $2|W|+1$, which are adjacent, so a word that is entirely a nesting is an ear and not a reduced core – its volume equals $p_T(W)$, as ear deletion requires. Nesting contributes reduced words only inside a concatenation, where the other factor separates the endpoints; that is exactly how ω_l and η_l are built.

Because evaluation is now a polynomial integral, the maximum over the whole class at a given number of pairs can be computed far past the range of any polytope enumeration. Doing so gives

$$r = 2 : \ \tfrac{2}{3}, \qquad r = 3 : \ \tfrac{1}{2}, \qquad r = 4 : \ \tfrac{9}{20}, \qquad r = 5, 6, \ldots, 16 : \ \tfrac{4}{9},$$

the small cases reproducing the constants of Corollary 9.3 and Theorem 9.1 and the rest agreeing with Proposition 4.1 at every size up to sixteen pairs. This is a maximum over the grammar class rather than over all reduced cores, and the class is thin: comparing against the complete reduced-core sweeps, it realises 3 of the 6 distinct volumes at $r = 5$, 6 of 19 at $r = 6$ and 8 of 46 at $r = 7$, so its share of the spectrum falls as r grows. Two things make the comparison worth making anyway. Every volume the class produces does occur in the true spectrum – 3/3, 6/6, 8/8 at those sizes, so the calculus introduces no spurious values – and at each of $r = 5, 6, 7$ the class maximum equals the global maximum. The evidence for $r \geq 8$ is therefore of the form: the extremum is known to lie in this class at every size where the comparison can be made, and inside the class it stays at 4/9 through sixteen pairs.

What lies outside is also identifiable, and it says what the calculus is missing. The cores realising the volumes the class does not reach are words such as *abacbdeced*, *abacbdcede* and *abcadbecde* at five pairs, or *ababcdefefcd* and *abacbdefcfed* at six: in each the cycles of the prefix-state cactus are glued at an interior shared state rather than at the base point. Concatenation joins blocks at the empty state and nesting places one inside a single pair; neither produces a gluing at an interior vertex, and for such a word the profile is not a product – the path-of-squares core *abacbdcd* is the smallest example, and it is exactly the case where the gap marginals differ from gap to gap (volumes $\frac{1}{4}$ and $\frac{313}{1260}$ for the same inserted word). An evaluation rule for gluing two cycles at an interior state would reach many of these; $\frac{313}{1260}$ duly appears among the six-pair volumes outside the present class. It would not reach all of them, because the block data of Lemma 7.1 is not a complete invariant for the volume. The words

$$abcdabcd, \qquad abacbdcd, \qquad abcadbcd$$

all have eight distinct prefix states on four coordinates, so each has a single cycle block of length eight using four labels – the same data in the classification – yet their volumes are

$$\tfrac{2}{5}, \qquad \tfrac{11}{30}, \qquad \tfrac{11}{30}.$$

What separates them is how the cycle sits in the cube: the first walks monotonically up and then down, the others zigzag. So a complete evaluation needs the embedded cycle, not merely its length and label set.

The other obvious coarse invariant fails too. Grouping all 105 order-eight pairings by their height sequence – the number of open chords after each slot, that is the Dyck path – gives 14 classes, and 13 of them contain more than one volume; the class of $(1, 2, 1, 0, 1, 2, 1, 0)$ carries 1, $\frac{2}{3}$ and $\frac{9}{20}$ at once. Note that *abacbdcd* and *abcadbcd* have different height sequences and the same volume $\frac{11}{30}$, while words sharing a height sequence routinely differ, so the Dyck path neither determines the volume nor is determined by it. The volume depends on which chord each down-step closes, not on the height profile – which is one reason these numbers have resisted a formula, and why the profile calculus buys what it does only on a structured subclass.

Among coarse statistics the crossing number does best. At order eight it takes seven values and five of the seven classes are pure; the failures are exactly at two crossings, where $\frac{1}{2}$ and $\frac{9}{20}$ both occur, and at three, where $\frac{1}{2}$, $\frac{11}{30}$ and $\frac{2}{5}$ all occur. The nesting number is nearly useless – six of its seven classes carry several volumes, and the class of nesting number zero already carries all six values – and refining to the pair (crossings, nestings) still leaves seven impure classes out of twenty-eight. That the crossing number is so nearly a complete invariant, and fails only in the middle of its range, invites the obvious refinement: the chord-intersection graph $G_\times(\pi)$ of Proposition 4.3 itself, rather than its edge count. This is the interlacement graph of the diagram, so the question is whether p_T factors through the circle graph.

It very nearly does. Grouping pairings by the isomorphism class of $G_\times(\pi)$ gives, at order six, four classes and all pure; at order eight, eleven classes and all pure, with $K_3 \mapsto \frac{1}{2}$ and $K_4 \mapsto \frac{2}{5}$ in agreement with $p_T(w_r) = 2/(r+1)$ for $G_\times(w_r) = K_r$, and the empty graph $\mapsto 1$. At order ten there are thirty-four classes and exactly one is impure: the graph on five vertices with two disjoint edges and one isolated vertex carries both $\frac{9}{20}$ and $\frac{4}{9}$. The witnesses are

$$aabcbcdede \;\mapsto\; \tfrac{9}{20}, \qquad ababcdedec \;\mapsto\; \tfrac{4}{9},$$

the second being ω_1. They have the same interlacement graph and differ only in what the fifth chord does: in ω_1 it nests the second crossing pair, in the other it is an ear nesting nothing. So the volume is not a function of the circle graph alone.

The failure is not arbitrary, and separating the classes by connectivity says where it lives. Through order ten there are twenty-nine isomorphism classes with $G_\times(\pi)$ connected — two at order six, six at order eight, twenty-one at order ten — and every one of them is pure. Of the twenty disconnected classes in the same range, nineteen are pure and the twentieth is the one displayed above, which is the only impurity anywhere through order ten. This is what the witnesses suggest: when $G_\times(\pi)$ is disconnected the diagram may realise its components either side by side or with one nested inside another, and the volume tells those apart, whereas a connected graph leaves no such freedom. We record the resulting question.

Conjecture 2.11. *If $G_\times(\pi)$ is connected, then $p_T(\pi)$ depends only on the isomorphism class of $G_\times(\pi)$.*

Remark 2.12. Part of this is classical. By Bouchet's theorem a circle graph that is prime for split decomposition has an essentially unique chord diagram, so for prime $G_\times(\pi)$ the conjecture is immediate — the graph determines the diagram, never mind the volume. Disconnection is the crudest way for primeness to fail, and it is the way our witness fails. What the order-ten data leaves open, and what would give the conjecture real content, is the connected but non-prime range: we have not checked which of the twenty-one connected classes are prime, and it is exactly the imprime ones that would make this more than a restatement.

The ingredient such a rule needs is a two-point kernel: the section volume with both the base point and the shared state held fixed. For the square it is elementary. The walk visits x, u_1, u_2 and $u_2 - u_1 + x$, so fixing x and $u = u_1$ and integrating out u_2 gives

$$K_{abab}(x,u) \;=\; \big(1 - |u - x|\big)_+,$$

and $\int_0^1 K(x,u)\,du \;=\; f_{abab}(x)$, $\iint K \;=\; \frac{2}{3}$, both checked exactly. Profiles are the one-point marginals of these kernels; concatenation and nesting only ever need the one-point data, which is why they close, while gluing at an interior state needs K itself.

The class also supplies a check of the calculus against the lattice geometry. Corollaries 5.2 and 5.3 say that the normalised volume $(r+1)!\,p_T$ of an r-pair core is an integer and, by the parity refinement, an even one. Those statements come from Gorenstein reciprocity and the mod-two h^*-identity; the grammar volumes come from integrating polynomials and never touch a polytope. All 628 volumes the class produces for $2 \le r \le 12$ have even integral normalised volume, with no exception.

The extremum is isolated, and by a fixed amount. The three largest volumes in the class are $\frac{4}{9}$, $\frac{1}{3}$, $\frac{3}{10}$ for every $r \ge 7$ (at $r = 5, 6$ the runners-up are $\frac{61}{180}$ and $\frac{1}{3}$), so the gap below the maximum is $\frac{4}{9} - \frac{1}{3} = \frac{1}{9}$ independently of r. This is a different phenomenon from the quantized improvement of Corollary 9.2, whose gap shrinks like $2/(r+1)!$: within this class the top of the volume spectrum does not accumulate at $4/9$ but sits a fixed distance below it.

The same computation runs on the Hankel side. Since w_r is parity mixing exactly for odd r, and both concatenation (an even shift) and nesting (which flips every parity) preserve parity mixing, the parity-mixing subclass is the same grammar with odd cycles only. Its maxima are

$$r = 3:\ \tfrac{1}{2}, \qquad r = 5:\ \tfrac{1}{3}, \qquad r = 6:\ \tfrac{23}{90}, \qquad r = 7, 8, \ldots, 18:\ \tfrac{1}{4},$$

reproducing Corollary 9.3 at three pairs, Proposition 10.2 at five, the six-pair maximum, and then the η_l value at every size up to eighteen. One honest gap: the grammar contains no four-pair parity-mixing word, so it misses the extremal 12314324 of volume $11/30$; the class is rich enough to carry the asymptotic extremals but not every small one.

The class already contains the extremal families: $abab\,cdcd$ gives $\frac{9}{20}$, the two-square chains ω_l give $\frac{4}{9}$, the w_3 chains η_l give $\frac{1}{4}$, and $w_3 w_3$ gives $\frac{23}{90}$. It also produces values that would be laborious otherwise, all confirmed against the exact engine: $abcabc\,dede$ gives $\frac{61}{180}$, $abab\,cdcd\,efgefg$ gives $\frac{13}{56}$, $abcabc\,dede\,fghfgh$ gives $\frac{166}{945}$, and $abcdbcda\,efghefgh$ gives $\frac{1}{5}$. In fourteen randomly generated words of the grammar the closed evaluation and the polytope volume agreed exactly in every case.

3. Exclusion lemmas for the limiting laws

Lemma 3.1. *Neither γ_T nor γ_H is a variance mixture of centered Gaussian laws.*

Proof. If $X = \sqrt{V}Z$ with Z standard Gaussian and $V \ge 0$ independent, then $m_4 = 3\,\mathbb{E}[V^2] \ge 3(\mathbb{E}V)^2 = 3m_2^2 = 3$. But $m_4(\gamma_T) = \frac{8}{3} < 3$ and $m_4(\gamma_H) = 2 < 3$. □

Lemma 3.2. *γ_H is not a scale mixture of semicircle laws.*

Proof. If $\gamma_H = \mathrm{law}(\sqrt{V}S)$ with S standard semicircular, then $\mathbb{E}[V^k] = m_{2k}(\gamma_H)/\mathrm{Cat}_k$. Here $\mathbb{E}[V] = 1$ and $\mathbb{E}[V^2] = m_4/2 = 1$, so V is degenerate at 1 and γ_H would be the semicircle law, contradicting $m_6(\gamma_H) = \frac{11}{2} \ne 5$. □

Lemma 3.3. *Neither law is a q-Gaussian. For γ_T, $m_4 = 2 + q$ forces $q = \frac{2}{3}$, and then $m_6 = 5 + 6q + 3q^2 + q^3 = \frac{287}{27} \ne 11$; for γ_H, $m_4 = 2$ forces $q = 0$, and $m_6 = \frac{11}{2} \ne 5$.*

Lemma 3.1 rules out any exact conditional-Gaussian or symbol-averaging representation of either law. Lemma 3.2 is consistent with the non-unimodality of γ_H: a scale mixture of semicircle laws is unimodal.

4. The global 4/9 bound: extremals and exhaustive evidence

Beyond the four-pair layer, Theorem 9.1's constant 9/20 is attained only at $k = 4$, and the natural question is the sharp constant for $r \geq 5$. The profile calculus supplies the extremal family in closed form.

Proposition 4.1 (The 4/9 family)**.** *For $l \geq 1$ let*

$$\omega_l \;=\; abab\, c_1c_2\cdots c_l\, dede\, c_l\cdots c_2c_1,$$

a reduced core with $r = l+4$ pairs (two square blocks joined by a chain of l nested pairs). Then, for every $x \in [0,1]$,

$$f_{\omega_l}(x) \;=\; \tfrac{2}{3}\, f_{abab}(x), \qquad \textit{hence} \qquad p_T(\omega_l) \;=\; \left(\tfrac{2}{3}\right)^2 = \frac{4}{9}.$$

Proof. The block $c_1\cdots c_l\, dede\, c_l\cdots c_1$ is a complete word nested under c_1; applying the flattening rule of Lemma 2.5(ii) l times reduces it to the constant profile $p_T(dede) = \frac{2}{3}$ (the two-pair crossing volume). The whole word is the concatenation of *abab* with that block, so profiles multiply by Lemma 2.1, giving $f_{\omega_l} = \frac{2}{3} f_{abab}$ and $p_T(\omega_l) = \frac{2}{3}\int_0^1 f_{abab} = \frac{2}{3}\cdot\frac{2}{3}$. □

Proposition 4.2 (Exhaustive verification for $r \leq 7$)**.** *Over all reduced Toeplitz cores with $r = 5, 6, 7$ pairs – that is* 293*,* 3326 *and* 44189 *labelled words,* 29*,* 196 *and* 1788 *dihedral orbits – the maximum volume is exactly* 4/9*, and it is attained on a single dihedral orbit at each r, namely the orbit of ω_{r-4}.*

Together these give the sharp form of the conjecture: for every $r \geq 5$,

$$p_T(\rho) \;\leq\; \frac{4}{9},$$

with equality exactly on the orbit of ω_{r-4} – an equality class that is nonempty for every $r \geq 5$, so the constant cannot be improved at any size.

The general-r statement reduces to a single finite slab lemma. Recall the proof of Theorem 9.1: reducedness forces $e \geq 2$; for $e = 2$ Corollary 7.2 gives 9/20 (only at $k = 4$) or 4/9; and for $e \geq 3$ the paper bounds by 9/20 or 2/5. Hence the global bound for $r \geq 5$ follows at once from the following strengthening of Lemma 8.2, which removes its criticality hypothesis.

Conjecture 4.3 (Three-slab bound)**.** *Let $L = X_1 + X_2 - X_3$ and let $M \neq N$ be slot forms, neither a coordinate nor $\pm L$. Then*

$$\Pr\big\{|L| \leq \tfrac{1}{2},\ |M| \leq \tfrac{1}{2},\ |N| \leq \tfrac{1}{2}\big\} \;\leq\; \frac{2}{5}.$$

By Lemma 2.5 every slot form has coefficients in $\{0, \pm 1\}$ with odd support of size at least three, so for each ambient dimension d this is a finite exact computation of the same kind as Appendix A and Theorem 10.4. For forms of minimal support the finiteness is absolute and the conjecture is a theorem.

Proposition 4.4 (Three triangles)**.** *Let M_1, M_2, M_3 be pairwise distinct slot forms of support three, none the negative of another. Then*

$$\Pr\big\{|M_i(X)| \leq \tfrac{1}{2},\ i = 1,2,3\big\} \;\leq\; \frac{2}{5},$$

with equality exactly for the star system of Lemma 4.6.

Proof. A support-three form with coefficient sum one has two entries $+1$ and one entry -1, so the configuration is a $3 \times d$ matrix over $\{0, \pm 1\}$ whose rows each carry two $+1$ and one -1, with no zero column. Permuting coordinates does not change the probability, so a configuration is determined by the multiset of its columns, each a nonzero vector in $\{-1, 0, 1\}^3$, and the row conditions say that for each row the multiplicities of the columns carrying $+1$ there sum to two and those carrying -1 sum to one. That integer system has exactly 463 solutions, of which 34 are degenerate (two forms equal or opposite). Computing the remaining 429 probabilities exactly gives maximum $2/5$, attained by five configurations, all equivalent to

$$M_1 = U + X_1, \quad M_2 = U + X_2, \quad M_3 = U + X_3, \qquad U = X_5 - X_4,$$

the star of Lemma 4.6 at $j = 3$. The next values attained are $3/8$, $11/30$, $17/48$, $61/180$ and $1/3$. □

Since the star is itself a triangle system, Proposition 4.4 settles the conjecture on the set where the extremum lives, and what remains is to rule out larger supports. Exhaustive scans over forms of every admissible support give maximum $3/8$ over all 55 pairs at $d = 4$, $2/5$ over all 741 at $d = 5$, and $2/5$ over all 7021 at $d = 6$; the value $2/5$ is Lemma 8.2's critical-triple constant. Since $\frac{2}{5} < \frac{4}{9}$, the conjecture yields

$$p_T(\rho) \le \frac{4}{9} \quad (r \ge 5), \qquad \text{equality exactly on the orbit of } \omega_{r-4},$$

a complete proof of the global bound modulo one finite certificate.

The Hankel analogue: a sharp $1/4$. The parity restriction forbids square blocks (Corollary 7.3), so the smallest cycle available in the parity-mixing class is $w_3 = abcabc$ with $p_H(w_3) = \frac{1}{2}$. Replacing squares by w_3 in Proposition 4.1 gives the Hankel analogue.

Proposition 4.5 (The $1/4$ family)**.** *For $l \ge 1$ let*

$$\eta_l \;=\; abcabc\, c_1 \cdots c_l\, def def\, c_l \cdots c_1,$$

a reduced parity-mixing core with $r = l + 6$ pairs. Then $f_{\eta_l} = \frac{1}{2} f_{w_3}$ and

$$p_H(\eta_l) \;=\; p_T(\eta_l) \;=\; \left(\tfrac{1}{2}\right)^2 = \frac{1}{4}.$$

Proof. Flattening (Lemma 2.5(ii)) applied l times reduces the nested block to the constant $p_T(w_3) = \frac{1}{2}$; concatenation multiplies profiles; and the Hankel value equals the Toeplitz value by Theorem 4.1, the words being parity mixing. □

Computing the complete reduced parity-mixing layers gives the exact maxima

$$r = 5 :\ \tfrac{1}{3}, \qquad r = 6 :\ \tfrac{23}{90}, \qquad r = 7 :\ \tfrac{1}{4}, \qquad r = 8 :\ \tfrac{1}{4},$$

the value at $r = 8$ being new here (the whole eight-pair layer is 4738 words in 198 orbits, 92 distinct volumes). The equality classes are also explicit: at $r = 5$ the cube cycle w_5; at $r = 6$ the single orbit of $w_3 w_3$, whose volume $\int_0^1 f_{w_3}^2 = \frac{23}{90}$ is again a concatenation value; at $r = 7$ exactly two orbits, the cube cycle w_7 and η_1; and at $r = 8$ the single orbit of η_2 (the cube cycle w_8 is not parity mixing – w_r is parity mixing precisely for odd r). Since η_l exists for every $r \ge 7$, this suggests the sharp Hankel companion of Proposition 4.1:

$$p_H(\rho) \;\le\; \frac{1}{4} \qquad (r \ge 7),$$

sharpening Corollary 10.5's $1/3$, which is attained only at $r = 5$. The two bounds have the same shape: $4/9 = p_T(w_2)^2$ on the Toeplitz side and $1/4 = p_H(w_3)^2$ on the Hankel side, each the square of the smallest admissible cycle's volume.

One family behind every slab constant. The separate constants appearing across Sections 8–10 are all values of the cube-cycle family. The fully crossing r-pair word w_r has exactly $r-1$ exceptional prefix-state forms, so writing $j = r-1$ for the number of slabs, Lemma 2.4 gives

$$p_T(w_{j+1}) \;=\; \frac{2}{j+2}.$$

Comparing with the constants in the paper:

j	$2/(j+2)$	where the constant appears
1	2/3	concentration of a three-uniform sum on a unit interval
2	1/2	Lemma 8.1, critical companions (equality)
3	2/5	Lemma 8.2, three critical slabs
4	1/3	Theorem 10.4, the four-slab theorem
5	2/7	the rank-five bound, open
6	1/4	predicted

Every constant matches, and by Lemma 2.4 the cube cycle attains it at each level. This suggests the single statement behind the whole hierarchy.

Lemma 4.6 (The star system attains $2/(j+2)$)**.** *Let $U = X_a - X_b$ for two coordinates $a \neq b$, and let $M_i = U + X_{c_i}$, $i = 1, \dots, j$, for distinct coordinates c_i outside $\{a, b\}$ (signs immaterial). Then*

$$\Pr\big\{|M_i| \le \tfrac{1}{2},\ 1 \le i \le j\big\} = \int_{-1}^{1} (1-|u|)^{j+1}\, du = \frac{2}{j+2}.$$

This is the exceptional system of the fully crossing word w_{j+1}, whose forms are $\Phi_0 + \Phi_{j+1} - \Phi_i$, $1 \le i \le j$.

Proof. U has density $(1-|u|)_+$. Given $U = u$, the constraint $|u + X_{c_i}| \le \frac{1}{2}$ puts X_{c_i} in an interval of length $(1-|u|)_+$, and the X_{c_i} are independent, so the conditional probability is $(1-|u|)^j$. Integrating against the density of U gives $\int_{-1}^{1}(1-|u|)^{j+1}du = 2/(j+2)$. □

Lemma 4.6 recovers every constant in the table by a single computation, and matches $p_T(w_{j+1}) = 2/(j+2)$ from Lemma 2.4, as it must. The exhaustive scan at $j = 3$ finds exactly this configuration at the maximum: the nine attaining pairs at $d = 5$ all have the form $L = X_0 + X_1 - X_2$, $M = -U + X_4$, $N = -U + X_3$ with $U = X_0 - X_2$.

Conjecture 4.7 (Slab hierarchy)**.** *For $j \ge 1$ distinct exceptional slot forms $M_1, \dots, M_j$,*

$$\Pr\big\{|M_i(X)| \le \tfrac{1}{2},\ 1 \le i \le j\big\} \;\le\; \frac{2}{j+2},$$

with equality exactly for the exceptional system of the fully crossing word w_{j+1}.

The cases $j = 1, 2, 4$ are theorems in the paper; Conjecture 4.3 is the case $j = 3$, and $j = 5$ is the open rank-five bound. Two reductions make each case finite. First, the coefficients are automatically in $\{0, \pm 1\}$ with odd support (Lemma 2.5). Second:

Lemma 4.8 (Private pairs may be deleted)**.** *Let $M_1, \dots, M_j$ be slot forms and suppose two coordinates $c \neq d$ occur only in M_1, with opposite signs there. Let M_1' be M_1 with those two entries deleted. Then*

$$\Pr\big\{|M_i| \le \tfrac{1}{2}\ \forall i\big\} \;\le\; \Pr\big\{|M_1'| \le \tfrac{1}{2},\ |M_i| \le \tfrac{1}{2}\ (i \ge 2)\big\},$$

and M_1' is again a slot form (its support drops by two, staying odd, and its coefficient sum is unchanged).

Proof. Write $Z = (M_1', M_2, \dots, M_j)$ and $W = \pm(X_c - X_d)$, so that the vector of forms is $Z + (W, 0, \dots, 0)$ with W independent of Z. The event is $Z + (W, 0, \dots, 0) \in K$ for the symmetric convex set $K = [-\frac{1}{2}, \frac{1}{2}]^j$. Now Z is a linear image of a uniform distribution on a cube, hence has a symmetric log-concave law, so Anderson's inequality gives $\Pr\{Z + v \in K\} \le \Pr\{Z \in K\}$ for every fixed v; averaging over $v = (W, 0, \dots, 0)$ gives the claim. □

The hypothesis of opposite signs is needed: deleting two private coordinates of the same sign would change the coefficient sum, so the reduced row is no longer a slot form and the induction does not close. What Lemma 4.8 does give is that an extremal configuration has no form carrying two private coordinates of opposite sign. A form may still carry several private coordinates provided they all have the same sign – a support-three form, for instance, can have its two +1 entries private – and such configurations are not reduced by the lemma. Combined with the exhaustive scans below, which settle every configuration living in dimension at most six, the cases left open are those in dimension at least seven in which each form's private coordinates all carry the same sign. Exhaustive checks: at $j = 3$ the maximum is $3/8$ over all 55 admissible pairs at $d = 4$ and $2/5$ over all 741 at $d = 5$, and the deletion step never increased the probability in 55 of 55 tested systems. At $j = 5$ the star system was confirmed to give exactly $2/7$, and a search over five-subsets of admissible forms found nothing exceeding it: exhaustively at $d = 5$ (maximum $29/120$) and by random sampling at $d = 6$ and $d = 7$ (maxima $37/160$ and $1891/10080$).

A sharper test than sampling is to ask whether the star can be improved locally. Starting from the star and replacing one of its forms by every other admissible slot form, computing each result exactly, gives at $j = 3$ a best value of $2/5$ over 111 replacements in $d = 5$ and over 351 in $d = 6$, and at $j = 4$ a best value of $1/3$ over 464 replacements in $d = 6$ and 1384 in $d = 7$ – 2310 exact recomputations in all, and in none of them does anything beat the star, which is therefore an exact local maximum under single-form moves, also when a spare coordinate outside its support is available. The $j = 4$ run is also a check on the machinery rather than on the conjecture, since Theorem 10.4 proves the bound $1/3$ there; a replacement exceeding it would have contradicted a proved theorem, and none does.

Those sampling figures should be read with care, and their own numbers say why. Simulating random j-tuples for $j = 4, 5, 6$ in dimensions up to $j + 4$ gives best observed values 0.301, 0.208 and 0.117 against bounds $1/3$, $2/7$ and $1/4$: the sampled maxima fall further below the bound as j and d grow, so a random tuple is almost never extremal-shaped and "no violation found" is weak evidence at large j. The weight is carried by the exhaustive scans in low dimension and by Proposition 4.4, which is a complete enumeration. What the simulations do confirm sharply is the equality case: the star system reproduces $2/(j + 2)$ to within Monte-Carlo error at every j from 2 to 7, including the values $1/4$ and $2/9$ beyond the range where the exact volumes were computed.

A route to the open case $j = 5$ already exists inside the paper's own apparatus. Theorem 10.4 proves $j = 4$ by writing $D = [-M \mid I_4]$, observing that its column matroid is regular of rank four, and invoking the rank-four classification ($M(K_5)$ restrictions plus $M^*(K_{3,3})$). The identical argument at $j = 5$ needs the rank-five regular matroid data, and the cosystole bounds of Douthitt, Israel and Kennard cited there cover regular matroids of rank at most six. So the rank-five $2/7$ bound looks accessible by the paper's existing method rather than a new one.

5. Unimodality: a moment certificate

Khinchin's theorem states that a symmetric law ν is unimodal about the origin if and only if ν is the law of UZ with U uniform on $[-1, 1]$ independent of Z; equivalently, if and only if

$$u_k \;:=\; (2k + 1)\, m_{2k}(\nu), \qquad k \ge 0,$$

is the moment sequence of a positive random variable (Z^2). This turns unimodality into an exactly checkable Hankel positivity condition.

Proposition 5.1 (Four-moment certificate for the Hankel law)**.** *γ_H is not unimodal. Indeed, with $u_0, \dots, u_4 = 1,\ 3,\ 10,\ \frac{77}{2},\ \frac{843}{5}$,*

$$\det \begin{pmatrix} u_0 & u_1 & u_2 \\ u_1 & u_2 & u_3 \\ u_2 & u_3 & u_4 \end{pmatrix} = \frac{815}{4} - \frac{1812}{5} + 155 = -\frac{73}{20} \ < \ 0.$$

Proof. If γ_H were unimodal, Khinchin's theorem would give $\gamma_H = \mathrm{law}(UZ)$ and hence $u_k = \mathbb{E}[(Z^2)^k]$, so the Hankel matrix $(u_{i+j})_{i,j\le 2}$ would be positive semidefinite. The displayed determinant, computed from $m_2 = 1$, $m_4 = 2$, $m_6 = \frac{11}{2}$ and $m_8 = \frac{281}{15}$, is negative. □

Non-unimodality of γ_H is known [1, Prop. A.2]; the point here is that the exact moments make it a four-line rational computation, and that the same test is informative for the Toeplitz law, where unimodality is open. Two controls show the criterion has teeth: for the semicircle law all computed minors are positive, and for the symmetric two-point law $\frac{1}{2}(\delta_{-1} + \delta_1)$ the criterion fails already at $D_1 = -4$.

Proposition 5.2 (Toeplitz unimodality is not obstructed through order 22)**.** *For γ_T the sequence $u_k = (2k+1)m_{2k}(\gamma_T)$ begins $1,\ 3,\ \frac{40}{3},\ 77,\ \frac{2724}{5},\ 4565, \dots$, and all six leading minors of (u_{i+j}) and all six of (u_{i+j+1}) are strictly positive. Hence no obstruction to the unimodality of γ_T is visible in the moments through order 22.*

This is consistent with, and weaker than, Conjecture 6.1: a scale mixture of semicircle laws is unimodal, so the semicircular-mixture conjecture implies unimodality of γ_T, and the two mixture tests indeed pass together at every computed order.

The status of the two laws is asymmetric here. For γ_H, non-unimodality is part of the original theorem of Bryc, Dembo and Jiang [1], who use the convexity form of the definition – a symmetric law ν is unimodal when $x \mapsto \nu((-\infty, x])$ is convex on $(-\infty, 0)$ – which is the form Khinchin's theorem characterises, so Proposition 5.1 is a certificate for exactly that statement. For γ_T, Sen and Virág [5] proved absolute continuity with density bounded by 8, answering part of the question raised in [1]; unimodality is not settled there and we are not aware of a later resolution. Proposition 5.2 says the moments through order twenty-two contain no obstruction to it.

6. Two conjectures

Conjecture 6.1 (Semicircular mixture)**.** *γ_T is a scale mixture of semicircle laws: there is a probability law μ on $(0, \infty)$ with*

$$\gamma_T = \mathrm{law}\big(\sqrt{V}\, S\big), \qquad V \sim \mu, \quad \mathbb{E}[V^k] = \frac{m_{2k}(\gamma_T)}{\mathrm{Cat}_k}.$$

The mixing moments through order 22 are

$$1,\ \ \frac{4}{3},\ \ \frac{11}{5},\ \ \frac{454}{105},\ \ \frac{415}{42},\ \ \frac{5960}{231},\ \ \frac{108573}{1430},$$
$$\frac{56067533}{225225},\ \ \frac{3065263961}{3403400},\ \ \frac{17293570111}{4849845},\ \ \frac{298983373961}{19492200},$$

and both Stieltjes–Hankel matrices built from them are positive definite: all six leading minors of (s_{i+j}) and all six of (s_{i+j+1}) are strictly positive. A finitely supported mixing law would annihilate a minor, so the evidence is nondegenerate at every available order. In generating-function form the claim reads $M_T(z) = \mathbb{E}[C(Vz)]$ with C the Catalan generating function, so

Theorem 6.4 expresses the primitive masses κ_r^T as explicit functionals of μ. The conjecture implies γ_T is unimodal. The mixing law is close to, but is not, a Gamma law: $\Gamma(3, \frac{1}{3})$ matches $\mathbb{E}V$ and $\mathbb{E}V^2$ and gives $\mathbb{E}V^3 = \frac{20}{9} \neq \frac{11}{5}$.

Conjecture 6.2 (Recurrence slope)**.** *Let b_k^2 be the Jacobi coefficients of γ_T, so that its orthogonal polynomials satisfy $p_{k+1} = xp_k - b_k^2 p_{k-1}$. Then*

$$b_{k+1}^2 - b_k^2 \;\longrightarrow\; \rho^2 \;=\; 0.6870697\ldots,$$

where $\rho = 0.8288582\ldots$ is the Sen–Virág operator-norm constant.

The exact values give $b_k^2 = 1,\ \frac{5}{3},\ \frac{7}{3}$ (increments exactly $\frac{2}{3}$), then increments 0.68952, 0.69342, 0.69200, 0.69103, through order 22: after the peak at $k = 4$ the sequence decreases monotonically toward ρ^2. The conjectured mechanism is that $b_k^2 \sim ck$ characterizes a Gaussian-type tail $\exp(-x^2/2c)$ for the limiting law, and the extreme eigenvalues of the normalized matrix then grow like $\sqrt{2c\log n}$, which is the Sen–Virág asymptotic $\rho\sqrt{2\log n}$ precisely when $c = \rho^2$. The conjecture is thus equivalent to the statement that the Gaussian tail variance of γ_T equals ρ^2.

The two conjectures are welded together by the following elementary but rigorous implication.

Proposition 6.3 (Increments control the mean volume)**.** *Let ν be a symmetric law with finite moments and Jacobi coefficients $b_j^2 > 0$ satisfying $b_{j+1}^2 - b_j^2 \to c > 0$. Then*

$$\Big(\frac{m_{2k}(\nu)}{(2k-1)!!}\Big)^{1/k} \;\longrightarrow\; c,$$

so Conjecture 6.2 forces the mean pairing-polytope volume $\mathbb{E}_\pi[p_T(\pi)]$ to decay at the exact geometric rate ρ^2.

Proof. The lattice-path expansion gives $m_{2k} = \sum_P \prod_i b_{h_i}^2$ over Dyck paths of semilength k, where $h_1, \ldots, h_k$ are the heights from which the down-steps fall. For $b_j^2 = j$ the sum is exactly $(2k-1)!!$: a down-step from height h chooses which of h open chords to close, so weighted paths biject with matchings of $\{1, \ldots, 2k\}$. Normalizing,

$$\frac{m_{2k}}{(2k-1)!!} \;=\; \mathbb{E}\Big[\prod_{i=1}^{k} \frac{b_{h_i}^2}{h_i}\Big],$$

the expectation over the open-chord process $(h_j)_{0\le j\le 2k}$ of a uniform random matching revealed left to right. This process is an explicit inhomogeneous Markov chain: after j reveals at height h, the next position closes an open chord with probability $h/(2k-j)$ (its partner is exchangeable among the $2k-j$ unrevealed positions) and opens a new chord otherwise.

Fix $\varepsilon \in (0, c)$ and choose H with $|b_h^2/h - c| \le \varepsilon$ for $h > H$; put $B = \max_{h\le H} \max(b_h^2/h,\ h/b_h^2) \ge 1$ and let L be the number of down-steps from heights $\le H$. Then

$$(c-\varepsilon)^k\, \mathbb{E}\big[\big(\tfrac{1}{B(c-\varepsilon)}\big)^L\big] \;\le\; \frac{m_{2k}}{(2k-1)!!} \;\le\; (c+\varepsilon)^k\, \mathbb{E}\big[\big(\tfrac{B}{c+\varepsilon} \vee 1\big)^L\big].$$

It remains to show $\mathbb{E}[x^L] = k^{O_x(1)}$ for every fixed $x \ge 1$, and $\mathbb{E}[y^L] \ge k^{-O_y(1)}$ for $y \in (0, 1]$; both follow from: $L \le \Lambda := \#\{j : h_j \le H\}$, and Λ is $O_H(\log k)$ with tails controlled at every exponential scale. Fix $x \ge 1$ and set $M = 4xH$. For $j \le 2k - M$ and $h \le H$ the chain steps down with probability $h/(2k-j) \le H/M = 1/(4x)$, so during a visit to $[0, H]$ before time $2k - M$ each step moves up with probability at least $1 - 1/(4x)$; the visit length is dominated by H plus a geometric variable whose x-th power has finite mean $g_x \le 2$ (since $x \cdot \frac{1}{4x} \le \frac{1}{4}$). The number of distinct visits is one plus the number of down-crossings of level $H{+}1$, dominated by independent indicators with success probabilities $(H{+}1)/(2k-j)$ and total mean $\le (H{+}1)\log(2k)$; the final M positions contribute at most $M = O_x(1)$ deterministically.

Hence $\mathbb{E}[x^\Lambda] \le x^{M+H}\exp\big((H+1)\log(2k)(x^H g_x - 1)\big) = k^{O_{x,H}(1)}$, and for $y \in (0,1]$, $\mathbb{E}[y^\Lambda] \ge y^{C_H \log k}\,\mathbb{P}(\Lambda \le C_H \log k) \ge k^{-O_{y,H}(1)}$ by the same estimates. Taking k-th roots and letting $\varepsilon \downarrow 0$ completes the proof. □

The next free-divisibility target, and why it is delicate. Equation (13.2) turns the order-sixteen conditional-positivity test into a threshold on m_{16}. With the moments now known through order twenty-two, that test and its successors at orders eighteen, twenty and twenty-two are all decided in the affirmative: for the odd block indexed by $\{1,3,5,7,9,11\}$ the Schur slack at order twenty-two is $\frac{2053548476285965691920969192182281957}{319351875377160931497786589380 0}\approx 6.43\times 10^5 > 0$. The first undecided test is the one whose new diagonal entry is r_{24}. Carrying out the same elimination on the even block indexed by $\{2,4,6,8,10,12\}$, and using $r_{24} = m_{24} - \frac{12916985465756056}{1091475}$, the condition $r_{24} \ge v^{\mathsf T} A^{-1} v$ becomes

$$m_{24}(\gamma_T) \;\ge\; \frac{5333852902522682067969148113707168133066912858 89}{360051624064032570884955002972066280 00} \;\approx\; 1.4814133713\times 10^{10}. \tag{1}$$

What makes this the sharp form of the question is how close the target is. The successive growth ratios m_{2k+2}/m_{2k} are

$$2.6667,\ 4.1250,\ 5.5030,\ 6.8557,\ 8.2065,\ 9.5639,\ 10.9292,\ 12.3010,\ 13.6771,\ 15.0556,$$

with second differences 1.3574, 1.3653, 1.3718, 1.3761, 1.3785, so linear extrapolation gives a next ratio near 16.44 and

$$m_{24} \;\approx\; 1.4821\times 10^{10},$$

which exceeds the threshold in (1) by about 0.045 percent, roughly one part in 2200. No extrapolation from eleven moments can resolve a margin that small, so the order-twenty-four test is genuinely undecided by anything short of the exact value; the sign of $m_{24} - (1)$ is a single rational comparison that settles whether γ_T fails free infinite divisibility at this order.

The moment is carried by its own top core class. Writing $r = k - j$ in the Catalan transform $m_{2k} = \mathrm{Cat}_k + \sum_{r\ge 2}\binom{2k}{k-r}\kappa_r$ turns the binomial into $\binom{2k}{j}$, so the summand is $T_j = \binom{2k}{j}\kappa_{k-j}$ and

$$\frac{T_{j+1}}{T_j} = \frac{2k-j}{j+1}\cdot\frac{\kappa_{k-j-1}}{\kappa_{k-j}}.$$

Under the scaling $\kappa_r \sim r!\,\gamma^r$ of Remark 2.8 this is $\frac{2k-j}{(j+1)(k-j)\gamma} \to \frac{2}{\gamma(j+1)}$ for j fixed: the transform is a Poisson-like sum concentrated at the boundary $r \approx k$. Summing the geometric-factorial tail gives the following.

Proposition 6.4 (Boundary-layer form of the transform)**.** *If $\kappa_r \sim r!\,\gamma^r$ with $\gamma = 2\rho^2$ as conjectured, then*

$$m_{2k} \;\sim\; e^{2/\gamma}\,\kappa_k \;=\; e^{1/\rho^2}\kappa_k, \qquad \frac{m_{2k}}{\mathrm{Cat}_k} \;\sim\; e^{1/\rho^2}\,\frac{\kappa_k}{\mathrm{Cat}_k}.$$

In particular the mixing moments of Conjecture 6.1 are the primitive masses up to the Catalan normalisation and one explicit constant, so identifying the mixing law is the same problem as determining the primitive-volume asymptotics.

The whole moment is thus carried by the single largest core class: the empty-core (semicircle) share of m_{2k} falls from 0.231 at $k = 4$ to 0.004 at $k = 8$, while the share from cores of size $r > k/2$ rises 0.460, 0.706, 0.735, 0.884, 0.896. The exact ratios m_{2k}/κ_k for $k = 2,\dots,8$ are

$$4.0000,\quad 5.5000,\quad 5.1011,\quad 4.9209,\quad 4.8068,\quad 4.7252,\quad 4.6630,$$

decreasing monotonically; first-order Richardson extrapolation gives 4.20–4.24 against the predicted $e^{1/\rho^2} = 4.2864$. This is a consistency check rather than a measurement: the extrapolation assumes a $1/k$ correction and uses only $k \le 8$. What it shows is that a constant obtained independently – from the Sen–Virág norm constant through Conjecture 6.2 – lands inside the extrapolation's own spread.

It also explains why finite-order expansions have not identified either law: every fixed-r term of the transform carries a Beta representation, hence contributes moments of a law supported in $[0, 1]$, whereas the true mixing law has unbounded support with an exponential tail. The unbounded part lives entirely in the boundary layer, which no fixed-order truncation sees.

A quantitative exclusion for D-finiteness. Guess-and-verify with held-out moments gives a size-explicit exclusion. For an ansatz with u unknown coefficients, impose the first u coefficient equations, solve exactly for the nullspace, and test every solution against the remaining known moments; with N known coefficients this is a genuine falsification test whenever $u \le N - 1$.

Using all twelve known Toeplitz coefficients (through order 22) and all ten Hankel ones (through order 18): every algebraic ansatz $P(z, M) = 0$ of bidegree up to $(6, 5)$ and every holonomic ansatz $\sum_{j\le 4} p_j(z)M^{(j)} = 0$ of degree up to 6 that is small enough to be falsifiable IS falsified – 13 algebraic and 12 holonomic sizes for γ_T, 11 and 10 for γ_H, with none consistent with the known moments. For instance, at order 14 the two uniquely determined ansatzes for γ_H predict 10894.88 and 10866.32 against the exact $m_{16} = \frac{9179819}{840} = 10928.356$.

This is evidence against D-finiteness of either series, not a proof; but it is consistent with the independent finding that a compressed recurrence for the volume-weighted series fails a low-order Wick check. Both series are factorially divergent – $m_{2k} \asymp (2k-1)!!\,\rho^{2k}$ under Conjecture 6.2 – so the natural analytic object is the Borel transform, and the unified conjecture reads: the Borel series $\sum_k m_{2k}\zeta^k/k!$ and $\sum_r \kappa_r \zeta^r/r!$ have radius of convergence $1/(2\rho^2)$.

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Conflicts of interest

Y. Shmalo holds equity interests in Heights Labs and Convex Nexus Capital.

Declaration of generative AI use

Generative-AI tools were used for language editing, organizational assistance, and checking algebraic presentation during manuscript preparation. The author independently reviewed the mathematical statements, proofs, and references and takes full responsibility for the content of the paper.

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